

The $k = 3$ Case of Conjecture 5.10

Theorem 1. *For every odd integer $n = 2m+1 \geq 7$, the almost-consecutive circulant tournament*

$$\text{AC}_n = \text{Cay}(\mathbb{Z}/n\mathbb{Z}, \{1, \dots, m-1\} \cup \{m+1\}),$$

where

$$i \longrightarrow j \iff j - i \pmod{n} \in g, \quad g = \{1, \dots, m-1\} \cup \{m+1\},$$

is $3\text{-}\vec{\omega}$ -critical.

Consequently, there are infinitely many $3\text{-}\vec{\omega}$ -critical tournaments, so the $k = 3$ case of Conjecture 5.10 of Aboulker–Aubian–Charbit–Lopes [1] holds. Moreover, Question 5.9 of [1] has a negative answer for $k = 3$.

Recall that the tournament clique number is

$$\vec{\omega}(T) = \min_{\sigma} \omega(B_{\sigma}(T)),$$

where the minimum is over all total orderings σ of $V(T)$, and $B_{\sigma}(T)$ is the back-edge graph: if $i <_{\sigma} j$, then $\{i, j\}$ is an edge of $B_{\sigma}(T)$ exactly when $j \rightarrow i$ is an arc of T . A tournament T is $k\text{-}\vec{\omega}$ -critical if

$$\vec{\omega}(T) = k \quad \text{and} \quad \vec{\omega}(T - v) = k - 1 \quad \text{for every } v \in V(T).$$

Proof. We divide the proof into four steps.

Step 1: AC_n is a tournament. For each $r \in \{1, \dots, m\}$, exactly one of r and $n - r$ belongs to g . Indeed, if $r \leq m - 1$, then $r \in g$, whereas

$$n - r = 2m + 1 - r \geq m + 2$$

does not belong to g . For $r = m$, we have $m \notin g$, while $n - m = m + 1 \in g$. Thus g contains exactly one element from each pair $\{r, -r\}$ of nonzero residues, so AC_n is a tournament. It is vertex-transitive, since every rotation $x \mapsto x + c$ is an automorphism. Notice that primality of n is not used.

Step 2: the upper bound $\vec{\omega}(\text{AC}_n) \leq 3$. Consider the identity order

$$0 < 1 < \dots < n - 1.$$

For $0 \leq i < j \leq n - 1$, the pair $\{i, j\}$ is a back edge if and only if $j \rightarrow i$, or equivalently

$$i - j \pmod{n} \in g.$$

In terms of the ordinary integer gap $j - i$, this is equivalent to

$$j - i \in D := \{m\} \cup \{m + 2, \dots, 2m\}.$$

In particular, every back-edge gap is at least m .

Suppose that

$$a_1 < a_2 < \dots < a_q$$

is a clique in the back-edge graph. Each consecutive gap is at least m , and therefore

$$a_q - a_1 \geq (q - 1)m.$$

On the other hand, $a_q - a_1 \leq n - 1 = 2m$. Hence $q \leq 3$, proving

$$\vec{\omega}(\text{AC}_n) \leq 3.$$

We also record the equality case. If $a_1 < a_2 < a_3$ is a back-edge triangle, then both consecutive gaps are at least m , while $a_3 - a_1 \leq 2m$. Consequently both gaps equal m , so

$$\{a_1, a_2, a_3\} = \{a_1, a_1 + m, a_1 + 2m\}.$$

Since $a_1 + 2m \leq 2m$, we must have $a_1 = 0$. Thus the unique back-edge triangle in the identity order is

$$\{0, m, 2m\}.$$

Step 3: the lower bound $\vec{\omega}(\text{AC}_n) \geq 3$. Let

$$N_0 = N^+[0] = \{0\} \cup g = \{0, 1, \dots, m-1, m+1\}.$$

Then $|N_0| = m+1$, and vertex-transitivity gives

$$N^+[v] = N_0 + v \quad \text{for every } v \in \mathbb{Z}/n\mathbb{Z}.$$

A pair $\{0, v\}$ dominates AC_n if and only if

$$N_0 \cup (N_0 + v) = \mathbb{Z}/n\mathbb{Z}.$$

Since $2|N_0| - n = 1$, this is equivalent to

$$|N_0 \cap (N_0 + v)| = 1.$$

No singleton dominates, because $|N_0| = m+1 < n$. The required domination lower bound therefore follows from the next lemma.

Lemma 1. *For every odd integer $n = 2m+1 \geq 7$,*

$$\min_{t \neq 0} |N_0 \cap (N_0 + t)| = 2.$$

Proof. Put

$$J = \{0, 1, \dots, m+1\}, \quad N_0 = J \setminus \{m\},$$

and let

$$K = (\mathbb{Z}/n\mathbb{Z}) \setminus J = \{m+2, \dots, 2m\}.$$

Thus $|J| = m+2$ and $|K| = m-1$. By taking complements,

$$|J \cap (J + t)| = n - |K \cup (K + t)| = 3 + |K \cap (K + t)|.$$

For $1 \leq t \leq m$, the set K is an interval of length $m-1$, and hence

$$|K \cap (K + t)| = \max\{0, m-1-t\}.$$

It follows that

$$|J \cap (J + t)| = \begin{cases} m+2-t, & 1 \leq t \leq m-1, \\ 3, & t = m. \end{cases}$$

Passing from J to $N_0 = J \setminus \{m\}$ removes m from the intersection whenever $m - t \in J$, which holds for every $1 \leq t \leq m$. It also removes $m + t$ whenever $m + t \in J$, which occurs exactly when $t = 1$. Therefore

$$|N_0 \cap (N_0 + t)| = \begin{cases} m - 1, & t = 1, \\ m + 1 - t, & 2 \leq t \leq m - 1, \\ 2, & t = m. \end{cases}$$

All these quantities are at least 2, since $m \geq 3$. Finally,

$$|N_0 \cap (N_0 + t)| = |N_0 \cap (N_0 - t)|,$$

so the calculation for $1 \leq t \leq m$ covers every nonzero residue t . The minimum is therefore 2. $\square$

For completeness, when $n = 7$, every nonzero t attains the minimum in Lemma 1. For every odd $n \geq 9$, the minimizers are exactly

$$\{m - 1, m, m + 1, m + 2\}.$$

Lemma 1 implies that no pair dominates AC_n , and hence

$$\text{dom}(\text{AC}_n) \geq 3.$$

Property 3.2 of [1] states that

$$\text{dom}(T) \leq \vec{\omega}(T) \leq \text{dic}(T)$$

for every tournament T . Consequently,

$$\vec{\omega}(\text{AC}_n) \geq 3.$$

Together with Step 2, this proves

$$\vec{\omega}(\text{AC}_n) = 3.$$

Step 4: criticality. By vertex-transitivity, it suffices to delete the vertex 0. The unique back-edge triangle in the identity order is $\{0, m, 2m\}$. Therefore the restriction of the identity order to $V(\text{AC}_n) \setminus \{0\}$ has a triangle-free back-edge graph, and so

$$\vec{\omega}(\text{AC}_n - 0) \leq 2.$$

Conversely, $\text{AC}_n - 0$ contains the directed triangle

$$1 \longrightarrow 2 \longrightarrow m + 3 \longrightarrow 1.$$

Indeed, the corresponding differences are

$$1, \quad m + 1, \quad 1 - (m + 3) \equiv m - 1 \pmod{n},$$

all of which belong to g . The vertices $1, 2, m + 3$ are distinct and nonzero, since $m \geq 3$. A tournament has clique number 1 if and only if it is transitive, so the presence of this directed triangle gives

$$\vec{\omega}(\text{AC}_n - 0) \geq 2.$$

Thus

$$\vec{\omega}(\text{AC}_n - v) = 2 \quad \text{for every } v \in V(\text{AC}_n).$$

This proves that AC_n is 3- $\vec{\omega}$ -critical.

Distinct odd values of n give tournaments of distinct orders. Hence

$$\{\text{AC}_n : n \geq 7 \text{ odd}\}$$

is an infinite family of pairwise nonisomorphic 3- $\vec{\omega}$ -critical tournaments.

It remains to state explicitly why Question 5.9 fails for $k = 3$. Every proper subtournament $A \subsetneq \text{AC}_n$ omits some vertex v , and therefore

$$A \subseteq \text{AC}_n - v.$$

By monotonicity under taking subtournaments,

$$\vec{\omega}(A) \leq \vec{\omega}(\text{AC}_n - v) = 2.$$

Thus the only subtournament of AC_n with clique number at least 3 is AC_n itself. Since n is unbounded, no bound $\ell(3)$ as in Question 5.9 can exist. $\square$

Relation to the result of Crew–Fan–Koerts–Moore–Spirkl

There is no contradiction with the 2026 result of Crew–Fan–Koerts–Moore–Spirkl [2]. Their theorem confirms Conjecture 5.8 of [1]: there exist functions f and ℓ such that

$$\vec{\omega}(T) \geq f(k) \implies T \text{ contains } X \text{ with } |V(X)| \leq \ell(k) \text{ and } \vec{\omega}(X) \geq k.$$

Question 5.9 asks for the strictly stronger statement in which the threshold function is the identity, namely $f(k) = k$. The tournaments AC_n all satisfy $\vec{\omega}(\text{AC}_n) = 3$, so they disprove this identity-threshold statement at $k = 3$. They do not conflict with a theorem whose hypothesis at $k = 3$ may require $\vec{\omega}(T) \geq f(3) > 3$.

Provenance and dependencies

The construction, the identity-order upper bound, the uniqueness of the back-edge triangle, Lemma 1, and the criticality argument are self-contained. The sole external input is Property 3.2 of [1], used to pass from $\text{dom}(\text{AC}_n) \geq 3$ to $\vec{\omega}(\text{AC}_n) \geq 3$.

References

- [1] P. Aboulker, G. Aubian, P. Charbit, and R. Lopes, *Clique number of tournaments*, arXiv:2310.04265, 2023.
- [2] L. Crew, X. Fan, H. Koerts, B. Moore, and S. Spirkl, *Characterizing large clique number in tournaments*, arXiv:2602.09863, 2026.